When Initial Success Becomes Useful Learning

Signal availability, transfer, and retention in behavior cloning followed by RL

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Status: revised research design, not experimental results. This proposal replaces the previous broad initialization sweep with a mechanism study. It retains Qwen2.5-0.5B-Instruct, arithmetic problems with 3/5/10 inputs, 15 evaluation attempts, pass@k/pass^k curves, and preempt-only Babel execution. All numerical training and resource allocations below are proposed settings to validate before the main experiments.

Abstract

A policy that occasionally succeeds on many problems provides more opportunities for group-relative reinforcement learning than a policy that always succeeds on a narrow subset. That observation does not establish that its updates will generalize or preserve prior successes. We study the distinction between **having a learning signal** and **taking a beneficial update**. Initial success probabilities determine how often GRPO receives mixed rewards; the directions of the resulting parameter updates determine which other problems improve or regress. We will test this account using a small exact model, matched-update interventions on a language model, and short online trajectories from two behavior-cloning checkpoints. The primary intervention matches observed success/failure composition while varying update alignment with a separate set of verified solutions. We then measure effects on disjoint solutions and independently sampled task success. A conditional retention experiment compares a standard gradient-projection control with update-norm-matched and sham controls. The contribution is a controlled empirical test of what success profiles fail to explain, not a new GRPO identity, a new gradient-projection algorithm, or a universal rule that intermediate success is optimal.

Recommended change in direction

The final remark in the supplied draft is the right motivation, but it is not yet a novel hypothesis. The project should ask: **at comparable initial success and reward-signal availability, when do updates transfer constructively, and when do they erode previously reliable behavior?** Difficulty-aware curricula, mixed-reward diagnostics, SFT diversity, and gradient conflict already have substantial prior work. The proposed study connects those observations through controlled comparisons and explicitly separates likelihood preservation from preservation of verified success.

1. What changes, and why

The earlier proposal asked whether initialization and online objective interact. That remains useful context, but a rank reversal between SFT checkpoints would mainly reproduce existing results [2, 4, 5]. Adding pass^k would improve measurement without, by itself, supplying a new mechanism. The new design gives priority to a manipulation that holds observed reward composition fixed and changes the direction of an update relative to other problems.

Three changes follow. First, **signal availability is a control variable and an established explanation**, not the headline discovery. Second, “spatial distribution” must have an operational meaning: which problems share helpful or conflicting parameter-update directions, not an unspecified position in a task space. Third, regression must be measured on the same problems using independent initial and final samples; an aggregate increase in pass^k cannot establish retention.

The phrase “best initial distribution” also needs an objective and a budget. A distribution favorable for finite-retry coverage need not favor repeated success. A distribution favorable for one-step gradient magnitude need not maximize held-out performance after many updates. We will not optimize over all possible initial policies or claim that the success histogram alone identifies an optimal initialization.

Scope decision

The core language-model matrix becomes **two BC checkpoints × two online objectives × three seeds = 12 trajectories**, supplemented by local interventions. The untouched model remains a baseline and suitability check, not a third online arm. The old continued-BC and equal-total-batch comparisons are deferred: adding BC steps to rollout rounds does not equate compute, and this allocation question would dilute the mechanism study. A retention intervention adds at most nine trajectories only after the core evidence justifies it. Restricted demonstrations, GSM8K, robotics, larger models, and broad optimizer sweeps are deferred.

This is an ambitious course project with a credible bounded contribution. It is not possible to promise a best-project award or conference novelty before measurement. A useful negative result would establish that readily measured gradient proxies fail to explain verified-success changes after reward composition is controlled.

2. Evidence audit and research position

The summaries below distinguish observations in a paper's setting from general claims. References are primary papers or official publication/model pages. Where only an official abstract was inspected, the table says so. Preprints are identified as such; unverified conference page numbers from the supplied BibTeX are not carried forward.

Research	What is established in the inspected source	Consequence for this project
DeepSeekMath, Shao et al. [1], Section 5.2 and Figure 7	Introduces GRPO and compares it with online rejection-sampling fine-tuning. Its GSM8K/MATH comparison at temperature 0.7 reports higher Maj@K without higher Pass@K.	This is a setting-specific observation, not a theorem that RL cannot expand coverage. Majority voting is not pass^k. Retain the two-objective control without claiming it is new.

Research	What is established in the inspected source	Consequence for this project
Quagmires, Kang et al. [2], Sections 2-4	Documents cases where SFT accuracy misranks subsequent RL outcomes and studies held-out loss and large-K pass rates as predictors.	A checkpoint rank reversal is prior art. Its results do not by themselves identify the mechanism of per-problem retention in our setting.
Pretraining to Post-Training, Shen et al. [3], Section 4 and Appendix H	In controlled chess models, analyzes amplification, tail discovery, wrong-mode amplification, and other top-k transitions, including loss. It also studies pretraining/RL scaling.	Do not claim that prior work ignores individual-state regression. Our outcome is repeatable verified trace success, and our intervention changes update geometry at matched reward composition.
Lightweight SFT, Li et al. [4], official ICLR abstract	Reports distributional forgetting and diversity-based checkpoint selection; proposes Adaptive Early-Stop Loss.	“Best SFT accuracy need not be best for RL” and diversity-aware preparation are already studied. We do not introduce another entropy stopping rule.
SFT Overtraining, Aphale and Liu [5], Sections 4.1-4.3	Links deeper SFT, entropy collapse, homogeneous rewards, and poor GRPO outcomes in the studied code-model settings.	More mixed groups as an explanation is not new. Test whether equally mixed groups can nevertheless have different transfer effects.
Understanding R1-Zero-Like Training, Liu et al. [6]	Examines base-model/prompt effects and optimization biases; motivates altered GRPO normalization.	Cite it for implementation and normalization, not as proof of a universal elicitation-only account. Freeze our loss explicitly.
CurES, Zeng et al. [7], Section 4	Analyzes prompt selection and rollout allocation through gradient optimization and variance; proposes an efficient curriculum.	Intermediate difficulty and gradient-based sampling efficiency are not unexplored. Our principal comparison holds observed reward counts fixed.
Advantage Symmetry, Yu et al. [8], preprint	Studies group/sample-level symmetry and difficulty adaptation; proposes asymmetric group-relative advantages.	We do not present medium-difficulty preference or symmetric reward coefficients as discoveries. We keep coefficients fixed in the main intervention.
Diverse SFT Traces, Zhang et al. [9], preprint	Studies verified route diversity and post-RL coverage; links preparation to mixed-reward opportunities.	A broad-versus-restricted demonstration experiment would overlap closely. Defer it; do not relabel it as our main novelty.
GRPO Dynamics, Mroueh [10], official ICML abstract	Analyzes binary-reward GRPO losses, normalization, regularization, and explicit policy-update forms.	Our elementary derivations are explanatory background and implementation checks, not a new general GRPO theory.
Plasticity and Stability, Qiang et al. [11], Sections 3-4 and official ICML record	Decomposes surrogate-reward and KL-regularization gradients and proposes probabilistic conflict resolution.	Neither “gradient conflict causes forgetting” nor projection for GRPO is a new central claim. We test a specific matched-evidence and same-task-retention question.
Gradient-Aligned Reward, Zheng et al. [12], preprint	Uses expert-anchor gradient similarity to shape rewards.	We use full trainable-adapter geometry as a diagnostic/intervention and retain binary rewards. Anchor alignment alone is not novel.

Research	What is established in the inspected source	Consequence for this project
F-GRPO, Plyusov et al. [13], preprint	Studies rare-correct trajectories missed by finite groups and mass redistribution; proposes difficulty-aware weighting.	Separate loss of alternative correct routes from loss of total success on a problem. Our verifier accepts every legal solution.
PCGrad and GEM [14, 15]	Develop gradient-conflict/projection approaches in multi-task or continual learning.	A projection arm is a known mechanism control, with explicit surrogate and stochastic-gradient limitations.

The specific proposed contribution is therefore narrow: **within fixed BC checkpoints, compare reward-composition-matched updates and test whether their alignment with independent solution gradients predicts and changes held-out success and retention.** Then test whether a surrogate-preserving intervention actually preserves sampled reliability, beyond simply shrinking updates. The closest methods address different objects: CurES optimizes prompt/rollout allocation [7]; Qiang et al. study reward-versus-KL gradient conflict [11]; GAR shapes rewards using a rollout and its own problem's expert anchor [12]. Here KL is absent, reward counts are matched, and the anchor and evaluation problems are disjoint from the source prompts. These distinctions define the experimental question; they do not prove an exhaustive first-work claim.

Corrections to the supplied framing

Use “probability,” not “odds,” for p(x). Do not infer pass^k from Maj@K. Do not interpret wrong-mode amplification as proof that a correct binary verifier directly rewards wrong outputs. With shared parameters, changes induced elsewhere can lower a problem's success probability; finite sampling, optimizer state, and objective weighting can also matter. This project tests one such account rather than assuming it explains Shen et al.'s observations [3].

3. Mechanistic framework

3.1 Success profiles describe outcomes, not update directions

For a fixed policy and stochastic decoder, let p(x) be the probability of a verified correct trace. Average first within an input-count stratum, then equally over the three strata. Define:

$$P_1 = \mathbb{E}_x[p(x)], \qquad C_k = \mathbb{E}_x[1-(1-p(x))^k], \qquad R_k = \mathbb{E}_x[p(x)^k]$$

These are pass@1, pass@k, and pass^k. For any positive integer k, C_k measures at least one success in k independent attempts; R_k measures all k attempts succeeding. They are dependent summaries of one success distribution. Reliability here means repeated-sampling consistency on an unchanged task, not adversarial robustness. The pass@K estimator and repeated-success interpretation follow established evaluation practice [18, 19].

A policy with p(x)=0.5 everywhere has P_1=0.5, C_k=1-(1/2)^k, and R_k=(1/2)^k. A policy perfect on half the problems and impossible on the rest has P_1=C_k=R_k=0.5 for every k. Neither policy is universally better. Moreover, a change in R_k can conceal which previously reliable problems were lost.

3.2 Reward variation creates an opportunity, not guaranteed progress

For G independent completions with binary rewards, a group is mixed if it contains a success and a failure. Its probability is:

$$M_G(x) = 1 - p(x)^G - (1-p(x))^G, \qquad \mathbb{E}_x[M_G(x)] = C_G - R_G$$

The training group size G and evaluation retry count k are distinct. We fix G=4 in ordinary training and vary evaluation k; the identity above evaluates C_k and R_k specifically at k=G. Changing evaluation k does not change the training gradients.

All-success and all-failure groups have zero relative reward advantages in the specified GRPO configuration. Mixed groups can produce a nonzero update, but cancellation can still occur. For Q total completions, expected mixed-group count is (Q/G) times the average mixed probability. Neither this identity nor the benefit of nondegenerate rewards is claimed as novel [5, 7, 8].

The missing quantity is direction. Let s(y,x) be the parameter gradient of the complete-trace log probability; let s+ and s- be its sample means over successful and failed traces in one group. If m of G traces succeed and q=m/G, then the unclipped on-policy group gradient, averaged over G traces, is:

$$\widehat{g}_x = \sqrt{q(1-q)}\,(\overline{s}_+ - \overline{s}_-), \qquad 0 < m < G$$

This expression assumes zero numerical epsilon and omits the common positive $1/L_{max}$ factor. With epsilon, replace the coefficient by q(1-q)/(sqrt(q(1-q))+epsilon). It follows by summing the standardized reward coefficients separately for successes and failures. Two groups with exactly the same m have the same coefficient but need not point in similar directions. The identity is a short derivation for this proposal, consistent with prior binary-reward gradient analyses [7, 10]; it is not a new convergence theorem.

Let d be the actual parameter displacement produced by an update, including Adam's preconditioning and momentum. For sufficiently small d, differentiability gives:

$$p_{\theta+d}(z) - p_\theta(z) = \nabla p_\theta(z)^\top d + O(\|d\|^2)$$

Thus a nonzero source update can hurt another problem when this inner product is negative. Euclidean cosine between raw gradients is not automatically the relevant quantity for Adam. We will inspect actual proposed displacements and will not infer transfer from gradient norms alone.

Different metrics also weight transfer differently:

$$\nabla R_k = \mathbb{E}_x[k p(x)^{k-1} \nabla p(x)], \qquad \nabla C_k = \mathbb{E}_x[k(1-p(x))^{k-1} \nabla p(x)]$$

An update can therefore improve retry coverage and reduce consistency, or the reverse. Varying k changes which success-probability regimes receive the greatest weight, so a method ranking can depend on k even when the underlying policies are unchanged. There is no single “most informative” initial distribution without specifying the target metric, task distribution, parameterization, sampling budget, and training duration.

3.3 An exact counterexample to success-profile sufficiency

Use two equally weighted binary tasks with reward 1 for action 1 and p_i=sigmoid(theta times phi_i). At theta=0, both tasks have p=0.5. Compare aligned features phi=(1,1) with opposing features phi=(1,-1). Initial P_1 and the entire C_k/R_k curves, as well as mixed-group probabilities, are identical. In the aligned model, positive theta improves both tasks. In the opposing model, one improves while the other deteriorates; mean success remains exactly 0.5 for every theta. For every k>1, its R_k nevertheless increases as the policy specializes, while C_k decreases; at k=1 both equal P_1. The corresponding full-batch mean-reward gradient at zero cancels in the opposing model.

This is an illustrative mathematical construction, not a claim about measured LLM behavior or a newly discovered interference theorem. It disproves a universal inference from initial success probabilities alone. Its role is to establish the missing variable and test the analysis code before expensive experiments. A real transformer has fixed shared representations, so the language-model intervention must test whether analogous geometry matters there.

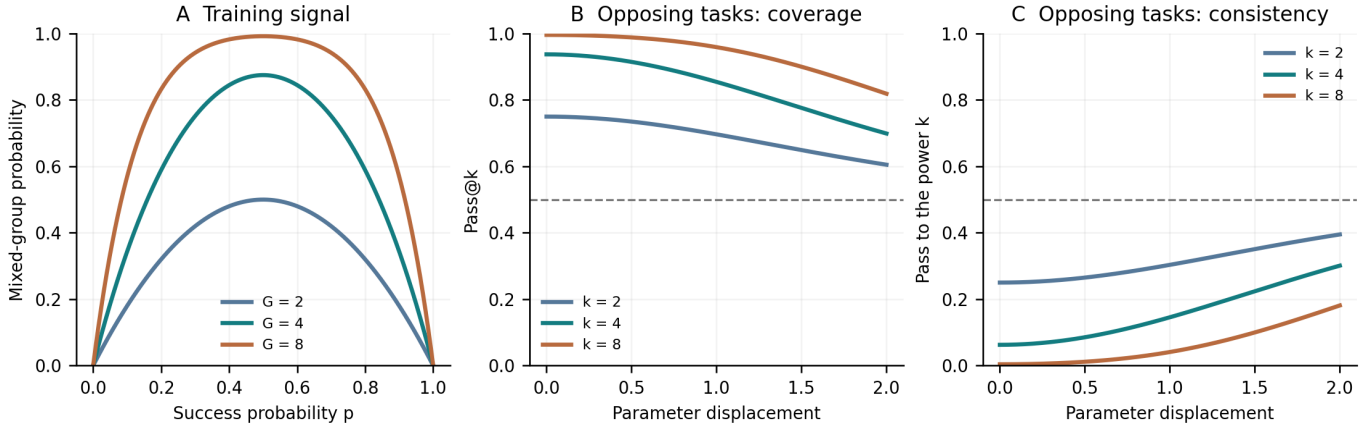


Figure 1. Analytic illustration, not measured results. Left: training-signal availability for several group sizes G. Middle and right: coverage and repeated-success curves in the opposing-feature example at illustrative evaluation k values. Specialization has different effects across metrics and k, despite unchanged average accuracy. Training G and evaluation k are separate quantities.

4. Three concrete hypotheses

Hypothesis and estimand	Experiment and predicted result	What would weaken or falsify it
H1: Reward composition is insufficient to predict transfer. At matched source reward counts, higher anchor alignment yields greater improvement on disjoint solution likelihood and, if measurable, verified success.	E2: fork identical checkpoints; apply matched-norm updates with high versus low anchor alignment. Primary contrast is the paired difference on an independent probe panel.	Alignment predicts only the anchor used for selection, not disjoint probes; or effects vanish under reward/length/norm matching. A likelihood-only result supports a surrogate claim, not success transfer.
H2: Initial successes can be lost through cross-problem updates even when direct relative reward signal on those problems is weak.	E3: track previously consistent held-out problems during online learning; relate losses to independently measured update effects, controlling descriptively for initial success, length, and stratum.	Reliable-group success remains stable, or losses follow truncation/verifier defects rather than update effects. Trajectory correlation alone cannot establish causality.
H3: A surrogate-preserving intervention can reduce verified retention loss beyond uniform update shrinkage.	Conditional E4: compare projected GRPO, displacement-norm-matched GRPO, and ordinary GRPO from the same extended-BC checkpoint.	Projection protects anchor likelihood but not held-out R_k, or gains disappear against norm matching, or preservation is bought by losing all online improvement.

H1 is the main experimental claim; H2 gives it relevance to the BC-to-RL trajectory; H3 is a bounded follow-up. A successful project need not confirm all three. We will not replace a failed hypothesis with an unregistered favorable metric.

5. Model, tasks, and data roles

5.1 Baseline-first suitability

Use Qwen2.5-0.5B-Instruct as the primary candidate. Its official report gives 49.6% GSM8K accuracy for the instruction-tuned 0.5B model [16], and its card documents the architecture [17]. Neither source establishes success under our arithmetic verifier, prompts, temperature, or output limit. Do not treat that score as our zero-shot result or derive pass^k from it.

Before BC or RL, evaluate 144 development problems, balanced across 3/5/10 inputs, with 15 attempts under both zero-example instructions and the fixed one-example format prompt. The latter is the main protocol. Seek useful headroom: at least one 5- or 10-input stratum should preferably have roughly 5-60% success, with valid outputs and non-ceiling coverage. These are engineering targets. Flag >80% initial accuracy as easy, <2% across assessed initializations as a floor, >20% format failure, or >5% truncation. Inspect examples and independent verifier checks before interpreting difficulty.

If necessary, allow one development-only generator revision, retaining 3/5/10 inputs, before freezing all splits. Examples include input magnitude or operator-mixture changes. A disposable light-BC model may distinguish format unfamiliarity from an unusable task. Keep 3-input problems

as an easy control and 10-input problems as a hard control when warranted. If no meaningful learning regime exists, report that result rather than tuning tasks until a preferred hypothesis holds. No automatic larger-model fallback is funded.

5.2 Verifiable arithmetic

Provide 3, 5, or 10 positive integer inputs and a target. A valid trace consumes two currently available operands, applies +, -, or multiplication, and returns the result to the multiset. Require exactly 2, 4, or 9 operations and one final value equal to the target. Accept all legal traces, not only the generator's solution. Reward is binary and terminal; prefix correctness and error labels are diagnostics only. Use exact integer arithmetic and a parser, never arbitrary code execution.

Generate solvable instances from binary expression trees with initial inputs 2-20, positive target at most 1,000, and intermediate absolute values at most 10,000. Apply the previous proposal's rejection rules uniformly; record acceptance rates, operator mix, and tree shape. The generating solution does not determine every possible solution's structure. Deduplicate by sorted input multiset and target. Separate operation count from token horizon. This is controlled multi-step sequence generation, not evidence about long robotic episodes.

Split	Size and use	Prohibited use
Development	144 problems plus 512 disposable BC demonstrations; suitability, implementation, rate/radius calibration	Final hypothesis evidence
BC	2,048 broad valid demonstrations, shared across seeds	Online evaluation/test labels
Online	4,096 broad prompts; uniform sampling for the main trajectories	Reference traces in ordinary GRPO/RFT updates
Anchor-fit	48 independent problems, 16/input count; verified traces define an alignment surrogate	Final test reporting; claiming its likelihood is task success
Anchor-check	48 separate problems; tests whether anchor effects generalize to other verified traces	Choosing high/low update pairs
Mechanism probe	48 separate problems; independent generated-success outcomes for local interventions	Selecting update pairs or radius
Validation	144 separate problems, 48/input count; intermediate trajectories and E4 gate	Selecting main claims from test outcomes
Locked test	288 problems, 96/input count; initial/final success and retention	Tuning data, learning rates, intervention, or stopping

All splits are disjoint by canonical key and shared across seeds. Seeds vary optimization and rollouts, not datasets; this does not estimate uncertainty over alternative dataset constructions. Reference traces in anchor sets are diagnostic extra supervision and must be disclosed and costed. They are not supplied to the ordinary online learner. The projection arm uses anchor-fit supervision explicitly.

6. Fixed training and measurement implementation

Train three BC paths with seeds 11, 22, and 33. Use effective batch 32 on 2,048 examples; save light BC at 16 steps (0.25 epoch) and extended BC at 128 steps (2 epochs). Stratify the first quarter by input count. These are two interventions, not guaranteed weak/strong policies or independently controlled reliability. Reset online optimizer state at each branch; preserve exact checkpoint hashes.

Use BF16 and LoRA rank 16, alpha 32, dropout zero, on attention q/k/v/o and MLP gate/up/down projections, with base weights frozen [21]. Record parameter names and counts. BC uses completion-token cross-entropy, AdamW 1e-4, weight decay zero, and gradient clipping 1. Online rate candidates are 5e-6 and 1e-5, calibrated equally on development and then frozen per objective. The actual one-step displacement, not the nominal learning rate, is used in alignment analysis.

For ordinary online learning, use 32 prompts × 4 independent completions = 128 completions per round. Sample at temperature 1, top-p 1, no top-k filtering, repetition penalty 1, with the same format example. Begin with a 256-token completion cap; raise the common cap to 512 before freeze if nine-operation solutions are truncated. Invalid and truncated outputs receive zero and remain in the denominator. All endpoints use the same frozen decoding procedure.

GRPO uses group mean and population standard deviation plus 1e-6; the objective is the negative clipped token-ratio surrogate summed over completion tokens and divided by Q times L_{max}. Use clip radius 0.2, KL zero, no critic, and exactly one accumulated optimizer step per fresh round. At the first and only step, ratios should be one, so clipping is inactive. This is a specified group-standardized, constant-length configuration, not the full Dr. GRPO recipe [1, 6]. Test the real implementation against hand-computed batches.

Filtered supervision retains successful fresh outputs S and minimizes their negative log likelihood summed over completion tokens and divided by |S| times L_{max}. It does not reuse earlier outputs. Skip a GRPO optimizer step when all advantages vanish and an RFT step when S is empty; still count the round and samples. This prevents momentum-only updates on empty learning signals. Report differing normalization scales rather than pretending the two objectives are equivalent.

Log prompt/trace IDs, outcomes, length, entropy, group success count m, gradient norm, actual displacement norm, skipped steps, processed/generated tokens, GPU type, elapsed allocation, and restart work. No reward shaping, success-based resampling, replay, or adaptive curriculum in the ordinary trajectories. Local intervention selection is a separate, explicitly conditioned experiment.

7. Experiments in execution order

E0. Exact model, verifier, and suitability

Implement the two-task logistic counterexample on CPU and verify the mixed-group formula, metric estimators, and Taylor approximation numerically. Sweep theta from -2 to 2 and G in {2,4,8}; this is an analytic control, not an LLM ablation. Check all-success/all-failure updates, operand multiplicity, negative intermediates, valid alternative traces, truncation, and checkpoint resume. Then run the baseline-first suitability study and freeze the task and resource forecast. Exit only when incorrect scoring and accidental model ceilings have been ruled out or documented as a stopping result.

E1. Establish initial profiles without overclaiming prediction

Produce the six BC checkpoints and the shared untouched-model baseline. Measure validation success profiles, mixed-group frequency, held-out BC loss, token entropy, and lengths. Before observing online outcomes, record expectations for each initialization. With only six learned checkpoints, do not fit a flexible readiness predictor or claim that one feature universally predicts RL. Use these profiles to describe the regime and to verify that the subsequent comparisons address different initial states.

E2. Matched-evidence update intervention: the primary experiment

For each of the six checkpoints, generate a fixed pool of at most 1,024 prompt groups from the online pool, four completions/group. Compute each group's GRPO gradient over all trainable LoRA parameters and the anchor-fit ascent direction h, defined as the gradient of mean constant-length-normalized log likelihood of its verified traces. Use streamed gradients/dot products; do not store a full parameter-by-group tensor or substitute a random projection without validation.

Construct up to eight disjoint pairs of minibatches, each minibatch containing eight mixed groups (32 completions). Within a pair, match the joint histogram of input count and m in {1,2,3} exactly; match mean completion length within 5%. The mixed fraction is therefore one in both arms, and observed successful-output counts are identical. Select groups to produce higher versus lower alignment with h, then calculate the actual candidate Adam displacements from identical copied optimizer states. Use those displacements to verify the contrast. The smaller local batch is intentional and common to both arms; it is not the main online batch size.

Rescale the two displacements to the smaller L2 norm within the pair. Calibrate one common radius multiplier on development so the median sampled token KL is small (target at most 0.01); freeze it before E2. Report achieved norms, KL, and alignment. Prefer opposite-sign pairs when available, but report how many exist. If only relative high/low contrasts exist, do not label the low arm harmful. Selection uses only anchor-fit data and source-pool statistics, never anchor-check or probe outcomes; freeze matching rules and tie-breaking seeds in advance. If matching fails, report the attained pair count and residuals; do not fabricate balance or resample indefinitely.

Apply each displacement once to a fresh clone of the same checkpoint. Never update sequentially across pairs. Evaluate the selection anchor, the independent anchor-check trace likelihood, and 15 fresh attempts per mechanism-probe problem. The primary *mechanistic* endpoint is the paired change in anchor-check likelihood; the primary *behavioral* endpoint is the paired change in probe pass@1, with the full pass@k/pass^k curves reported alongside it. A likelihood contrast by itself cannot establish improved reasoning or reliability. Separate format-validity gains from

arithmetic-validity gains in the failure analysis; report likelihood on numerical/operator versus delimiter tokens as a diagnostic so template imitation is not mistaken for reasoning transfer. One-step behavioral effects may be below the resolution of this panel, which is an admissible negative outcome.

This intervention identifies the effect of the specified batch-selection and displacement-normalization procedure. It does not independently randomize a universal property called geometry: source problem identities and latent difficulty can still differ. Matching observed m also does not guarantee equal true p(x). Report these limits. Holding a checkpoint fixed does, however, eliminate initialization weights, evaluation distribution, optimizer state, reward coefficients, and update norm as explanations of the paired contrast.

As a sensitivity analysis, report paired sign-flip results only with their exchangeability assumption stated. High/low batches are selected, not randomly assigned: this is not an exact design-based causal randomization test. Treat checkpoints and pairs as clustered units rather than hundreds of generated traces as independent trained models. The analytic Taylor check on the selection anchor is an implementation check, not empirical novelty.

E3. Online trajectories: does the mechanism matter beyond a single update?

Run two initializations × {GRPO, filtered supervision} × three seeds = 12 branches. Propose 256 rollout rounds per branch, with validation at rounds 0, 64, 128, and 256. Use 15 attempts at the initial/final validation endpoints and four at intermediate monitoring points; intermediate curves are lower-precision diagnostics, not substitutes for final 15-attempt estimates. Freeze the final round count after development throughput checks; all arms receive the same sampling opportunity. There are 32,768 completions per full branch. Count skipped rounds, not only successful optimizer steps.

Record the qualification rule and subset sizes at initialization; do not redefine the reliable group after seeing regressions. At fixed validation checkpoints, compute actual update displacements and anchor-fit/check likelihood changes on a prespecified audit batch. Track learning-signal availability and displacement effects separately. Test whether a high mixed-group fraction can coexist with negligible improvement or adverse transfer. Associations along a trajectory are descriptive; E2 supplies the more controlled intervention.

Measure retention using independent initial samples: use the first seven attempts to label observed consistency (six or seven successes), and the remaining eight to estimate an independent initial baseline. Final estimates use all 15 fresh attempts. For initial-to-final retention comparisons, restrict k to 1-8 because the independent initial baseline has eight samples. Final levels at k=9-15 may be shown descriptively, but cannot support a matched initial-to-final retention claim. Analyze the same labeled problems at the final checkpoint. Report how many qualify by input count; if fewer than 12 qualify in a stratum, show counts and descriptive outcomes without a strong stratum-level retention claim. The grouping defines observed consistency, not true p=1.

The main contrast is GRPO versus filtered supervision within initialization and seed, followed by the difference of those method contrasts across initializations. Report all seed points. Initial success, entropy, and held-out loss are prespecified descriptive comparators, not covariates for a high-dimensional significance search. Full test outcomes are opened only after final model and analysis choices are fixed.

E4. Conditional retention control: projection versus smaller updates

Run this only if final validation pass@1 on the initially consistent subset falls by at least five percentage points in at least two extended-BC seeds, each with at least 12 qualifying problems, scoring/truncation errors do not explain it, and the resource forecast permits completion. This prespecified practical trigger is not a significance claim; report R_k and uncertainty as well. Freeze that gate before examining the locked test. If it fails, report “not run: mechanism not established or resources insufficient,” not a negative intervention result.

From each extended-BC checkpoint, add three arms: projected GRPO, displacement-norm-matched GRPO, and a sham projection using a fixed random unit direction in adapter parameter space. Compare with the ordinary GRPO runs already in E3: nine additional trajectories total. All arms use the same round budget and initial prompt streams; their completions naturally diverge after updates.

For the projection arm, compute h as the current anchor-fit log-likelihood ascent gradient and d as the proposed Adam displacement. When h dot d is negative, use:

$$d_{\text{proj}} = d - \frac{\min(0, h^\top d)}{\|h\|^2} h$$

Use d unchanged when h has negligible norm and log the event. This is a known first-order projection construction [14, 15], not a new optimizer. Apply it to the displacement after computing the Adam proposal; keep the Adam moment update from the unprojected gradient and document this design. It preserves the linearized anchor-likelihood surrogate, not true task success or finite-step monotonicity. Recompute h each round and count the extra forward/backward work.

For the norm control, compute the same anchor projection counterfactually on that arm's own d, then scale d by ||d_proj||/||d|| without changing direction. This controls shrinkage while giving the arm comparable diagnostic overhead; it is not a cheaper production baseline. Sham projection controls an arbitrary directional restriction. Compare initial-to-final retained-subset R_k curves for k=1-8 and total $P_1/C_k/R_k$ curves for k=1-15, as well as norm, KL, and anchor likelihood. A benefit over ordinary GRPO but not the norm control supports a step-size explanation, not the proposed retention mechanism. An anchor-only benefit reveals surrogate failure. Report which k ranges support a retention benefit and which remain unresolved; do not summarize one favorable k as uniform reliability improvement.

8. Statistical design and metric feasibility

Use **pass@k and pass^k as metric families**, not a single favored k. Prespecify all integer k from 1 through 15 for full-panel evaluation, and display complete curves with compact tables at k={1,2,4,8,15}. At k=1 both families equal pass@1. Keep pass@1 as the primary scalar behavioral outcome; coverage and consistency claims concern their prespecified curves. Larger k means more chances to find one success for pass@k, but a stricter all-attempts requirement for pass^k. These are different operational questions, not alternative estimators of the same quantity.

For n independent attempts on a problem with c successes, estimate:

$$\widehat{C}_k = 1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}, \qquad \widehat{R}_k = \frac{\binom{c}{k}}{\binom{n}{k}}, \qquad 1 \le k \le n$$

Define choose(a,k)=0 for a<k. Use n=15 for full initial/final evaluation and average at problem level, then equally across input counts. The same 15 outputs produce every k value without extra model sampling; these estimates are correlated. The combinatorial estimators avoid the nonlinear plug-in bias of substituting c/n into population formulas [18]. Do not estimate k>15 from these outputs without extra samples or an explicitly validated model. Intermediate four-attempt monitoring supports only k≤4. The independent eight-attempt initial retention baseline supports only k≤8; final samples can still use n=15 at those same k.

The k ablation tests claim robustness. Report whether method rankings persist, reverse, or become unresolved as k changes. A gain at small k and a loss at larger k supports a budget-dependent tradeoff, not uniformly improved coverage. A high-k floor is a measurement limit, not equality. Do not choose the k with the largest observed advantage after viewing results. No area-under-curve composite will replace the separate outcomes.

Fifteen attempts do not solve the rare-event problem. For homogeneous p, true repeated success is p^k and the chance of a nonzero per-problem estimator is Pr(Binomial(15,p)≥k). At p=0.1, R_2=0.01, R_4=0.0001, and R_8=0.00000001; the corresponding nonzero-estimate probabilities are about 45.10%, 5.56%, and 0.00336%. At k=15, the reliability estimator is simply an indicator that all 15 trials succeeded. High-k comparisons will often be unresolved, even when low-k effects are informative. These are analytic examples, not Qwen performance forecasts.

If every reliability score is zero at a particular k, use one prespecified k-subset per problem for a conservative rare-event bound. For N independent identically sampled problems and zero all-success events, the one-sided upper bound is 1-alpha^(1/N). To cover all k=2-15 and three strata simultaneously, use alpha=0.05/(14*3) in each bound; report the resulting conservative uncertainty rather than a zero-width bootstrap interval. Other reported families need their own explicitly declared multiplicity accounting.

Use 10,000 paired whole-problem bootstrap resamples, retaining all attempts and stratifying by input count. For curve-level claims, construct simultaneous paired bootstrap bands across the prespecified k grid within each reported metric/comparison family, using the resampled maximum absolute estimation error; label pointwise intervals separately. If claiming across strata or multiple arm contrasts, include those dimensions in the family. Degenerate rare-event endpoints need the bounds above rather than nominal bootstrap certainty. Show seed-level results separately; three seeds do not justify precise claims about training-seed variability [20]. For E2, the update pair is also a repeated experimental unit: report within-checkpoint pair differences and checkpoint/seed summaries. Shared probe problems and shared starting weights mean that fork outcomes are not independent training runs. Distinguish uncertainty over problems, selected batches, and seeds.

The 288-problem test is not a promise of fine resolution. For a bounded per-problem score, a worst-case normal-approximation 95% half-width is about 5.8 percentage points overall and 10 points per 96-problem stratum, before accounting for seed variability. Pairing may improve precision. The 48-problem E2 probe is substantially less precise; its likelihood endpoint is sensitive, but its behavioral null cannot rule out small effects. Report effect sizes and intervals, not only whether a p-value crosses a threshold.

Initial success groups use independent baseline samples to avoid regression-to-the-mean artifacts. Report full-panel R_k curves and initial-to-final curves on initially consistent problems over their supported k grids; these answer different questions. “Newly observed success” does not mean a capability was absent initially. Alternative correct-route diversity is a secondary diagnostic, not a substitute for total success probability [13].

9. Compute, deadline, and operational plan

9.1 Fixed scientific exposure; revisable resource forecasts

GPU-hours are resource accounting, not matched scientific exposure across unlike hardware. Training boundaries are BC steps/epochs and online rollout rounds. Report generated and processed tokens, active update counts, allocated time, GPU type, and preemption rework. Prefer matched GPU classes within pairs, but do not give slower GPUs fewer rounds merely to match hours. Do not equate a BC batch, an online round, and a unit of FLOPs.

The proposed **core envelope is 240 allocated GPU-hours**, to be revised using measurements before the main matrix. It replaces the earlier forecast because gradient interventions add work. This is not a claim about measured throughput or an owner-specified total grant. The conditional extension needs a separately documented forecast and must not displace the core report.

Component	Proposed GPU-hours	What is included
Development, profiling, and correctness	20	Baseline suitability, disposable BC, loss and restart checks, calibration
Main BC paths	12	Three paths with two saved checkpoints
E2 local interventions	36	Candidate generation, streamed gradients, fork construction, likelihood diagnostics; generation-based evaluation charged below
E3 online training	84	Twelve branches; about seven hours/branch is a planning average, not a stopping rule
Evaluation and analysis	58	Initial/final metrics, E2 probes, intermediate validation, loading and verification
Reserve	30	Preemption, required recomputation, and evaluation shortfall
Core total	240	Revise with evidence; preserve report and controls

If E4 is authorized by its gate and schedule, a provisional additional 100 GPU-hours covers nine training branches (63), extra evaluation/anchor-gradient work (25), and reserve (12). Reprofile this separately: its 38,880 additional locked-test completions and per-round anchor gradients are not free. This extension is not included in the 240-hour core forecast.

The core final-test manifest has seven initial policies (shared untouched model plus six BC checkpoints) and twelve final policies: $19 \times 288 \times 15 = 82{,}080$ completions. E2 has at most 96 one-update forks plus six starting checkpoints: $102 \times 48 \times 15 = 73{,}440$ mechanism-probe completions. Combined, these are 155,520 completions, at most 39.8 million output tokens with a 256-token cap, or 79.6 million with 512. Initial/final/intermediate validation adds 52,704 completions under the stated schedule, and the initial two-prompt development audit adds 4,320, bringing these specified evaluation workloads to 212,544 completions. Further calibration is additional. E2 candidate generation adds at most 24,576 completions. E3 generates 393,216 training completions at 256 rounds. Include all of these in the measured forecast.

Using 58 GPU-hours for the 212,544 specified evaluation completions requires about 261 generated tokens/second at the 256-token cap, before additional calibration and overhead, or 521 at 512. These are arithmetic planning requirements, not attainable-throughput claims. Actual mean lengths may be shorter. Gradient passes, model loading, prefix processing, idle allocation, and verification must be measured. A short canary on real hardware determines whether the proposed schedule is credible.

9.2 Babel constraints and completion sequence

Retain the project's hard pool and ownership limits: no general GPUs, no auto-routing, no ae-topup, and no interference with other projects. The supplied snapshots imply six uncommitted preempt slots after the two-GPU buffer, but the other ae study retains a ninth reserved slot; start with five BC2RL GPUs. Verify live shared-user allocations, pending requests, reservations, and effective QOS limits before submission; lower usage when necessary. Pending jobs, evaluations, canaries, and tinkers all count. A sixth slot requires the documented release already specified in PROJECT.md.

Use one process/GPU, project-isolated harness paths and job IDs, CPU-only work on preempt_cpu_qos, and no GPU standby. Save atomic checkpoints at least every ten minutes, including adapter/optimizer/RNG/sampler states and cumulative counters. Test restart before a fleet. Monitor through deterministic events; keep state/checklist current and publish progress/timeline updates every four hours. Only BC2RL-owned jobs may be moved or cancelled, within permitted pools.

The current requested reporting target is Tuesday, 27 October 2026, 09:00 EDT. Immediately reconcile the existing repository and any already-running experiments before adopting this revised plan. Preserve valid prior results, label their protocol, and do not present them as new matched-intervention runs. Do not discard running work merely because the proposal changed.

Execution priority is E0, E1, E2 and the twelve E3 branches, then final evaluation and report. E4 is conditional and may follow the first report. If the deadline forecast is poor, freeze a shorter common round budget before the main matrix and retain the mechanism/control structure. If jobs still remain incomplete, deliver an explicitly partial report on time with usable artifacts and exact missing comparisons. No automated resource revision may violate the hard concurrency limits.

10. Outcomes, decision rules, and deliverables

Observation	Supported conclusion	Unsupported conclusion
Matched reward counts, different independent likelihood and success effects	Reward availability alone does not explain local transfer under this intervention	A universal readiness predictor or causal geometry independent of source identity
Alignment predicts anchor/check likelihood but not verified success	The proxy captures trace likelihood but may fail as a behavioral mechanism	Preserved reasoning competence
Mixed groups increase but final success does not	More opportunities are insufficient in this regime	Mixed groups are irrelevant to GRPO
R_k rises overall at some k while the initially consistent subset deteriorates	Aggregate repeated success at that k can hide retention loss	The policy is uniformly more reliable
Projection beats ordinary GRPO but not norm matching	Smaller updates are a plausible explanation	Directional protection is necessary
Projection beats both controls on retained R_k with useful overall gains	Evidence for the tested retention intervention in this setting	A new projection method or universal prevention of wrong-mode amplification
No local or sustained behavioral effect is resolved	The mechanism is unsupported or too small for this design's resolution	Exact equality of methods or proof that geometry never matters

Deliver a report with the frozen protocol, source-to-claim audit, full run matrix including failures, input-count-stratified metrics, all seed results, local matched-pair diagnostics, retention analysis, costs, and claim limitations. Supply reproducible scripts, configuration hashes, dataset manifests, verifier tests, checkpoint provenance, and attempt-level outputs. Keep large data/checkpoints on compute storage and harvest small summaries through the existing harness.

The principal figures are: initial success profiles; mixed-group opportunity versus measured improvement; paired high/low update effects; per-problem initial-to-final success with independent grouping; retention versus overall coverage; and, if run, projection versus norm/sham controls. Show metric-versus-k curves by input count, initialization, and objective, including difference curves and uncertainty. A supplementary three-axis plot must identify its chosen k and show other prespecified k values in separate panels; it does not identify the mechanism.

Before submitting the final empirical paper, perform a technical/evidence review and a separate writing/figure review. Independently recompute sampled estimator and resource rows. Distinguish analytic examples, proposed settings, observed results, and inferences. The deliverable from this proposal stage is the design; no experimental outcome is asserted here.

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Sources checked for this revision on 24 October 2026. Primary-paper URLs are provided for reproducibility. Abstract-only evidence is explicitly labeled above. Bibliographic entries omit unverified pagination; preprint status does not establish peer review.

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Comprehensive ablation and control table

Core means funded in the proposed first study; **conditional** requires the E4 gate and a revised forecast; **fixed** means a control, not a sweep; **diagnostic** means measurement or a bounded check. This table describes experimental choices, not results.

Experimental knob	Values / setting	Status	Purpose	What the comparison reveals or cannot reveal
BC exposure	0.25 versus 2 epochs; untouched model for baseline only	Core	Create two initial states at one architecture	Initialization-dependent transfer; not independently randomized reliability
Training seeds	11, 22, 33; paired branches	Core	Expose optimization /sampling variability	Consistency across three seeds, not precise seed-population inference
Online objective	GRPO versus fresh success-filtered supervision	Core E3	Relative reward credit versus positive-only learning	Objective interaction with initialization; not algorithm-family superiority
Input count	3, 5, 10; 2, 4, 9 operations	Core strata	Compare controlled task complexity	Operation-count association; token length and branching also vary
Baseline prompting	Zero examples versus one fixed format example	Diagnostic E0	Separate prompting/format effects from task familiarity	Zero-shot evidence only for the zero-example condition
Task revision	At most one development-only generator revision	Conditional E0	Avoid a ceiling or universal floor	Revised task must be frozen; no selection for a preferred ranking
Token cap	256; 512 only if development truncation warrants	Fixed after E0	Permit valid long traces without gratuitous generation	Cap sensitivity is an engineering check, not a post-test sweep
Local alignment	Higher versus lower anchor alignment	Core E2	Manipulate update transfer at fixed checkpoint	Effect of a specified selection procedure, not pure geometry isolated from all task identity
Local reward composition	Match joint input-count and m=1/2/3 histograms	Fixed E2	Hold observed success/failure signal constant	Removes reward-count explanation; does not equate true p(x)
Local update norm	Both displacements scaled to smaller pair norm	Fixed E2	Remove update-size explanation	Directional contrast under this normalization
Local length balance	Pair means within 5%; publish residuals	Fixed E2	Reduce completion-length confounding	Remaining distributional differences still limit attribution
Local batch / pool	8 groups × 4; at most 1,024 candidate groups/checkpoint	Fixed E2	Bound gradient and generation costs	Local intervention differs from ordinary 32-group online learning

Experimental knob	Values / setting	Status	Purpose	What the comparison reveals or cannot reveal
Local replication	Up to 8 disjoint batch pairs/checkpoint	Core E2	Assess batch-to-batch variability	Forks share weights and probe problems; not independent training seeds
Local radius	One development-calibrated multiplier; target sampled token KL ≤0.01	Fixed E2	Keep the local approximation useful	Does not guarantee detectable success changes
Anchor roles	Fit 48, check 48, behavior probe 48; disjoint	Fixed	Separate selection, surrogate transfer, and verified behavior	Anchor-only improvement cannot establish reasoning preservation
Format versus reasoning	Validity errors; numerical/operator versus delimiter likelihood	Diagnostic	Detect template imitation	Format-only gains are not evidence of arithmetic transfer
Geometry representation	Full trainable LoRA displacement/gradient	Fixed	Match the actual optimization subspace	No claim about full-parameter training or invariant geometry
Main round budget	Proposed 256; validation at 0/64/128/256	Fixed after E0	Equal online sampling exposure	Equal rounds are not equal FLOPs or time across objectives
Main group size	G=4, 32 prompts, 128 completions	Fixed	Keep signal coefficient and exposure comparable	No group-size optimization claim; G=8 LLM sweep deferred
Analytic group size	G=2/4/8 on the CPU exact model only	Diagnostic	Verify signal formulas	Not a measured LLM result
Projection control	Actual-displacement projection onto anchor half-space	Conditional E4	Test surrogate-preserving direction changes	Existing construction; cannot guarantee task-level retention
Norm control	Rescale each arm's own proposal to its counterfactual projected norm	Conditional E4	Isolate shrinking updates	Projection must beat this control for a directional explanation
Sham control	Fixed random unit anchor direction	Conditional E4	Control arbitrary directional restriction	Helps distinguish meaningful anchor information from generic constraint
Initial consistency	First 7 attempts: ≥6 successes; baseline uses other 8	Diagnostic	Prevent grouping/baseline reuse	Observed consistency, not true p=1; report small subsets
Evaluation attempts	15 independent trials/problem for all compared policies	Fixed	Estimate the C_k/R_k families from common outputs	Correlated estimates; high-k R_k can be unresolved
Evaluation k	Full grid 1-15; tables at 1/2/4/8/15	Core evaluation ablation	Test retry-budget and consistency-order dependence	Report rank reversals and floors; no post-hoc best-k selection

Experimental knob	Values / setting	Status	Purpose	What the comparison reveals or cannot reveal
Retention / monitoring k	Retention 1-8; intermediate monitoring 1-4	Fixed measurement limits	Respect independent initial and monitoring sample counts	No extrapolated initial retention estimate beyond k=8
Training G versus evaluation k	Training G=4; evaluation k varies independently	Fixed distinction	Separate learning-signal mechanism from evaluation	Changing k does not test a different GRPO group size
Primary / secondary outcomes	E2 independent likelihood and P_1; E3 P_1 plus C_k/R_k; E4 retained R_k plus overall outcomes	Prespecified	Separate mechanism, behavior, and retention	No post-hoc composite or favorable-metric substitution
BC batch / optimizer	32; AdamW 1e-4, zero weight decay, grad clip 1	Fixed	Define exposure and implementation	Not a supervised-optimizer comparison
Online rate	5e-6 or 1e-5 per objective, equal development calibration	Calibration	Avoid numerical or scale failures	No per-initialization tuning or claim of optimal rates
Loss normalization	Constant L_{max} denominator; group std retained for GRPO	Fixed	Avoid inverse-response-length weighting	Not the full Dr. GRPO recipe
KL / critic / replay	None in ordinary GRPO; one optimizer step per fresh batch	Fixed	Limit mechanisms under study	No general claim about regularized or multi-epoch PPO variants
Reward	Exact binary full-trace correctness	Fixed	Avoid reward-model errors and reward shaping	Does not remove shared-parameter interference
Model and adapters	Qwen2.5-0.5B-Instruct; LoRA r16/alpha32/dropout0	Fixed	Keep a feasible common policy class	No model-scale or full-fine-tuning generality
Decoding	Temperature 1, top-p 1, top-k off, repetition 1	Fixed	Define a common success probability	Reliability depends on this stochastic decoder
Data construction	Fixed broad tree generator; canonical disjoint splits	Fixed	Control leakage and task weighting	No resampled-training-dataset uncertainty estimate
Resource envelope	Proposed core 240 GPU-hours; measured revision before release	Planning	Preserve feasibility and evaluation	Hours are accounting, not cross-hardware equal compute
Deferred scope	GSM8K, robotics, route-diversity intervention, no-BC online arm, compute-allocation optimizer	Not in core	Protect the mechanism study	Cannot claim cross-domain generality or optimal BC/RL allocation